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Experiment 2

Course Outcome (CO1): Describe the basic concepts of machine learning
BTL: L3
Objective: Perform various operations on the dataset using NumPy for numerical computation.
Dataset: Iris Dataset

A. Dataset Import and Exploration

1. Import the Iris dataset and convert the numerical data into a NumPy array.
2. Display the dimensions (number of rows and columns) of the dataset.
3. Determine the total number of elements in the numerical dataset.
4. Display the data type of the NumPy array.
5. Display the first 10 and last 10 records of the dataset.
6. Separate the feature data from the target/species information.
7. Display the names of all features present in the dataset.
8. Determine the number of samples and features in the dataset.

B. Array Indexing and Slicing

9. Extract the values of the first feature for all flowers.
10. Extract the first 20 samples from the dataset.
11. Extract the first three features from all samples.
12. Extract the last two features from the dataset.
13. Extract every alternate row from the dataset.
14. Extract the values corresponding to samples 10 to 30.
15. Extract a subset containing the first 10 samples and first 2 features.

C. Mathematical Operations

16. Calculate the mean value of each feature.
17. Find the minimum and maximum value of each feature.
18. Calculate the range of each feature.
19. Calculate the variance of each feature.
20. Calculate the standard deviation of each feature.

D. Comparison and Conditional Operations

21. Identify all samples having sepal length greater than 6 cm.
22. Count the number of samples having petal length greater than 5 cm.
23. Identify samples where sepal width is less than 3 cm.
24. Find samples where both petal length is greater than 4 cm and petal width is greater than 1.5 cm.
25. Count the number of samples satisfying a given numerical condition.

E. Using iloc and loc

26. Select the first 10 rows using iloc.
27. Select sepal_length and sepal_width using loc.
28. Select rows 20–30 using iloc.
29. Select rows with petal_length > 5 using loc.
30. Select the first 3 rows and first 3 columns using iloc.